A C_4-Free Classification of Square Relations on Token Graphs

with girth, tree and discretization witnesses

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Abstract

Let $F_K(X)$ denote the K-token graph of a finite simple graph X: vertices are the K-subsets of $V(X)$, adjacent when their symmetric difference is an edge. Two families of 4-cycles in $F_K(X)$ arise naturally when one attaches 2-cells: the *commuting squares*, given by two token moves along base edges with disjoint closed supports, and *all* 4-cycles of the token graph. Paper I of this series showed these coincide for cycle substrates C_n with $n \geq 5$. We prove the governing condition is neither a cycle hypothesis nor a girth hypothesis:

$$\mathcal{K}_4(F_K(X)) = \mathcal{K}_{\text{disj}}(F_K(X)) \iff X \text{ has no subgraph isomorphic to } C_4,$$

for $2 \leq K \leq |V(X)| - 1$, the bound being necessary since the token graph degenerates to a point at $K = |V|$. The forward direction is a short argument on closed token walks; the converse combines the token-graph complementation $F_k \cong F_{n-k}$ with an explicit spectator construction and an exhaustive check of the finitely many small cases. Consequently girth at least 5 collapses the triangle, square and mixed relation families to one, and the bull graph — 4-cycle-free but containing a triangle — shows the triangle and square conditions are logically independent. We exhibit a 7-vertex tree whose token graph carries first cohomology of dimension 2, so configuration topology is not inherited from the base cycle space. Finally we compute the collapsed invariant on girth-≥5 substrates and partition the results by whether the discretization theorem licenses a continuous reading: at $K = 2$ the improved subdivision bound is what makes the identification legitimate for the Petersen graph, the Heawood graph and the tree, while the original bound fails for all three.

Keywords: token graph, graph configuration space, graph braid group, discrete Morse theory, cube complex, relation layer

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1. Introduction and relation to Paper I

Paper I [1] studies invariant first cohomology of a token graph relative to a *declared* family of 2-cells, and treats that family as a variable rather than as background. Among its results is a lemma stating that for the cycle

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substrate C_n with $n \geq 5$, every 4-cycle of the token graph is a commuting square. That lemma is correct. It is also a special case of something that mentions no cycles.

This paper isolates the general statement, proves it in both directions, and records the consequences. The results here are stated in the language of token graphs alone and do not depend on the relation-layer apparatus of [1]; readers interested only in configuration spaces of graphs may read this paper independently. Where a quantity from [1] is used — the invariant residue $\varepsilon_{\mathrm{harm}}$ — it is defined in §2 for completeness.

The route to the general statement is worth recording because it was not a generalization found by inspection. The narrow lemma was tested against twenty substrates chosen so that they *could* violate it. Several did, and the pattern of failures identified the correct hypothesis: the bull, butterfly and paw satisfy the conclusion while containing triangles, ruling out girth; trees and paths satisfy it while containing no cycles at all; and K_4, K_5, $K_{3,3}$, Q_3 and C_5+chord fail it, every one of them containing a 4-cycle.

2. Setting

Definition 2.1. (token graph; Fabila-Monroy et al. [2]) For a finite simple graph X on n vertices and $K \geq 1$, the token graph $F_K(X)$ has as vertices the K-subsets of $V(X)$, with S adjacent to T whenever their symmetric difference is a pair of adjacent vertices of X. Equivalently, T is obtained from S by moving one token along one edge of X to an unoccupied vertex.

Definition 2.2. (the two square families) A *commuting square* is a 4-cycle of $F_K(X)$ of the form $S \to (S\setminus\{u\})\cup\{w\} \to (S\setminus\{u,v\})\cup\{w,x\} \to (S\setminus\{v\})\cup\{x\} \to S$, where uw and vx are edges of X whose closed supports are disjoint. The closed support of an edge uv is the vertex set it spans, written $S(uv) = \{u, v\}$; the condition is $S(uw) \cap S(vx) = \emptyset$. Write $\mathcal{K}_{\mathrm{disj}}$ for the set of these and $\mathcal{K}_4$ for the set of all 4-cycles of $F_K(X)$ on four distinct vertices. Always $\mathcal{K}_{\mathrm{disj}} \subseteq \mathcal{K}_4$.

Commuting squares are exactly the 2-cells of Abrams' discretized configuration space $UD^K(X)$, whose d-cells are d base edges with pairwise disjoint closed supports together with $K - d$ parked tokens [3]. Attaching $\mathcal{K}_{\mathrm{disj}}$ therefore builds the 2-skeleton of that complex; attaching $\mathcal{K}_4$ builds something strictly larger whenever the two families differ. We write $\varepsilon_{\mathrm{harm}}(\mathcal{K})$ for $\dim H^1$ of the 2-complex obtained by attaching $\mathcal{K}$, computed over $\mathbb{Q}$, and $\varepsilon_{\mathrm{harm}}(\mathcal{K} \mid G)$ for its G-invariant analogue as in [1].

3. The classification

Theorem 3.1. Let X be a finite simple graph on n vertices and let $2 \leq K \leq n - 1$. Then $\mathcal{K}_4 = \mathcal{K}_{\mathrm{disj}}$ in $F_K(X)$ if and only if X has no subgraph isomorphic to C_4, not necessarily induced.

The qualification is essential and is the likeliest source of a misreading. K_4 contains four-cycles as subgraphs while containing none as *induced* subgraphs, and it does *not* satisfy the equality: at $K = 2$ it has 15 token-graph 4-cycles of which only 3 are commuting. A reader testing the criterion with induced squares would wrongly record K_4 as a counterexample. Throughout, "contains a 4-cycle" means contains a subgraph isomorphic to C_4.

Proof of sufficiency. Assume X is 4-cycle-free and let (A, B, C, D) be a 4-cycle of $F_K(X)$ on four distinct vertices. Traversing the loop, each token describes a closed walk in X, and the walk lengths sum to 4. No token walk has length 1, and a length-3 walk would leave a residual length of 1, so the possible partitions are $\{4\}$ and $\{2, 2\}$.

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Suppose the partition is {4}. One token makes a closed walk of length 4 while the others are parked. Write the walk as $v_0 \to v_1 \to v_2 \to v_3 \to v_0$ and let P be the parked set. The four token-graph vertices visited are $\{v_i\} \cup P$, so they are pairwise distinct precisely when v_0, v_1, v_2, v_3 are pairwise distinct. In that case the four consecutive base edges exhibit a subgraph isomorphic to C_4, contrary to hypothesis. If instead $v_i = v_j$ for some $0 \le i < j \le 3$, the corresponding token-graph vertices coincide, contradicting the assumption that the cycle has four distinct vertices. Thus this case cannot occur.

Suppose the partition is {2, 2}. Two tokens each traverse one base edge and return, say along uw and vx. A closed walk of length 2 traverses one edge and immediately traverses it back, so each moving token uses a single base edge twice. If one token completed its out-and-back before the other moved, the token-graph vertex at that point would equal the starting one and the cycle would lack four distinct vertices; so the two moves alternate, giving the cyclic order of Definition 2.2. The base edges must have disjoint closed supports, since a shared vertex would have to be unoccupied for one move and occupied for the other at the same step. Hence the 4-cycle is a commuting square. ■

Proof of necessity. Assume X contains a 4-cycle on distinct vertices w, x, y, z. We produce a 4-cycle of $F_K(X)$ that is not a commuting square.

First reduce to $K \le n/2$. Complementation $S \mapsto V(X)\backslash S$ is an isomorphism $F_K(X) \cong F_{n-K}(X)$, since A and B have symmetric difference an edge precisely when their complements do; this is equation (2) of [2]. The isomorphism carries commuting squares to commuting squares, because two tokens moving along disjoint base edges correspond to two holes moving along the same two disjoint base edges. Hence $\mathcal{K}_4 = \mathcal{K}_{\text{disj}}$ holds at K if and only if it holds at $n - K$.

Now suppose $n \ge 6$ and $K \le n/2$. Then $n - K \ge n/2 \ge 3$, so $n \ge K + 3$ and at least $K - 1$ vertices of X lie outside $\{w, x, y, z\}$. Let P be a set of $K - 1$ such vertices and put $S = \{w\} \cup P$. The sequence

$$\{w\}\cup P \;\to\; \{x\}\cup P \;\to\; \{y\}\cup P \;\to\; \{z\}\cup P \;\to\; \{w\}\cup P$$

is a 4-cycle of $F_K(X)$ on four distinct vertices, since w, x, y, z are distinct and P is fixed and disjoint from them. Every one of its four steps moves the *same* token, whereas a commuting square is by Definition 2.2 built from two *distinct* tokens each making one out-and-back move. So it is not a commuting square. ■

The remaining cases are $n = 4$ and $n = 5$, where no spectator set of the required size exists. These were verified exhaustively by enumeration over all labeled simple graphs on four and five vertices containing a 4-cycle — 10 and 476 graphs respectively — at every K with $2 \le K \le n - 1$. No graph satisfied $\mathcal{K}_4 = \mathcal{K}_{\text{disj}}$. This is a finite computation and is stated as such; it is not covered by the spectator argument above.

> **Remark 3.2. (the bound on K is necessary)** At $K = n$ the token graph is a single vertex with no edges, so $\mathcal{K}_4 = \mathcal{K}_{\text{disj}} = \emptyset$ and the equality holds vacuously for every substrate, including graphs with many 4-cycles. The hypothesis $K \le n - 1$ excludes exactly this degeneracy.

4. Girth at least 5, and the independence of the two conditions

A triangle-free substrate admits no 3-cycle in its token graph: an arena 3-cycle requires three K-sets pairwise differing by one move, and whether or not a token is parked the three moves use the edges of a base triangle [1, Lemma 6.1]. Combining that with Theorem 3.1:

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Corollary 4.1. If girth(X) ≥ 5 and $2 \le K \le n-1$, then $\mathcal{K}_3 = \emptyset$ and $\mathcal{K}_4 = \mathcal{K}_{\text{disj}}$; hence $\mathcal{K}_{\text{disj}} = \mathcal{K}_4 = \mathcal{K}_{3,4}$ and the three corresponding residues agree.

The two hypotheses are independent, and a single substrate separates them. The **bull** — a triangle with two pendant vertices — contains no 4-cycle, so Theorem 3.1 gives $\mathcal{K}_4 = \mathcal{K}_{\text{disj}}$. It does contain a triangle, and at $K = 2$ under its full automorphism group:

$$\varepsilon_{\text{harm}}(\mathcal{K}_{\text{disj}}) = \varepsilon_{\text{harm}}(\mathcal{K}_4) = 3, \quad \varepsilon_{\text{harm}}(\mathcal{K}_{3,4}) = 0.$$

So 4-cycle-freeness alone delivers the square collapse and not the full collapse. A control of this kind is more informative than a further positive example: it isolates which clause is doing the work rather than accumulating cases where both hold.

5. Trees: configuration topology is not inherited

A natural but false heuristic holds that a substrate with no cycles, or with large girth, should have token graphs with no interesting first cohomology. Corollary 4.1 makes that heuristic tempting, since it says high girth simplifies the relation families. It is wrong about the cohomology.

Proposition 5.1. Let Y be the 7-vertex tree with edges 01, 12, 13, 34, 35, 56. Then $H^1(Y; \mathbb{Q}) = 0$, while $\varepsilon_{\text{harm}}(\mathcal{K}_{\text{disj}}) = 2$ for $F_2(Y)$ and 4 for $F_3(Y)$, computed at the trivial group.

The token graph $F_2(Y)$ has 21 vertices and 30 edges; $F_3(Y)$ has 35 and 60. The base graph is acyclic, so its cycle space is zero, yet the token graph has loops: with two tokens on a tree, passing one another requires routing one into a side branch while the other goes by, and the resulting closed configuration path is not null-homotopic in the complex of commuting squares. These loops are collision-avoidance motions around essential vertices; they need not be token exchanges, and on a tree with several branch vertices the independent classes come from distinct branch points. Collision avoidance creates the loops that the base lacks.

This is a known phenomenon in graph configuration spaces rather than a new one — it is the reason graph braid groups of trees are nontrivial — but it is worth stating explicitly here, because within the relation-layer framing it blocks any attempt to read residual cohomology off the base cycle space.

6. The collapsed invariant, and what the discretization theorem licenses

On a girth-≥5 substrate all the declared families coincide, so there is a single number to compute. At the trivial group that number is dim H^1 of the 2-skeleton of $UD^K(X)$, and first cohomology of a CW complex depends only on cells through dimension two, so it is dim $H^1(UD^K(X); \mathbb{Q})$. Measured:

X	girth	K	token graph	$\varepsilon\text{harm}(\mathcal{K}\text{disj}) = \varepsilon\text{harm}(\mathcal{K}_{3,4})$
C_5	5	2	10v 15e	1
C_7	7	2	21v 35e	1
Petersen	5	2	45v 120e	6
Petersen	5	3	120v 420e	6
Heawood	6	2	91v 252e	8
tree Y	∞	2	21v 30e	2

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X	girth	K	token graph	εharm($\mathcal{K}$disj) = εharm($\mathcal{K}_{3,4}$)
tree Y	∞	3	35v 60e	4

Table 1. The collapsed invariant at G = 1. Each displayed value equals dim $H^1(UD^K(X); \mathbb{Q})$, since the commuting-square complex is the 2-skeleton of $UD^K(X)$ and first cohomology is determined by cells through dimension 2. §6.1 decides separately which rows may be read as continuous configuration-space cohomology.

6.1 The subdivision hypothesis decided per graph

A vertex of *X* is *essential* if its degree is at least 3; this is the convention of [3, 4], under which a path or cycle has none. A loop is *homotopically essential* if it is not null-homotopic in *X*.

Abrams' theorem [3] gives a deformation retraction of the configuration space onto its discretization when (A′) every path between distinct essential vertices has length at least $K + 1$, and (B′) every homotopically essential loop at a vertex has length at least $K + 1$. Prue and Scrimshaw [4] weaken (A′) to $K - 1$ by discrete Morse theory, leaving (B′) unchanged.

For cycle substrates there are no essential vertices, so (A′) is vacuous and the improvement changes nothing; Paper I notes this correctly for that family. For every other substrate in Table 1 the improvement is decisive, and the original bound fails. All of Petersen, Heawood and *Y* have adjacent vertices of degree at least 3, so their minimum essential-vertex path length is 1:

X	K	min ess. path	Abrams (A′)	Prue–Scrimshaw (A′)	(B′)	continuous reading
Petersen	2	1	fails (needs 3)	holds (needs 1)	holds	yes — via [4] only
Petersen	3	1	fails (needs 4)	fails (needs 2)	holds	no
Heawood	2	1	fails (needs 3)	holds (needs 1)	holds	yes — via [4] only
Heawood	3	1	fails (needs 4)	fails (needs 2)	holds	no
tree Y	2	1	fails (needs 3)	holds (needs 1)	vacuous	yes — via [4] only
tree Y	3	1	fails (needs 4)	fails (needs 2)	vacuous	no

Table 2. The hypothesis checked per graph. Where the last column reads no, the Table 1 value remains a correct computation of discrete configuration-space cohomology and is not promoted further.

Accordingly: dim $H^1(\mathrm{UConf}_2(\text{Petersen}); \mathbb{Q}) = 6$, dim $H^1(\mathrm{UConf}_2(\text{Heawood}); \mathbb{Q}) = 8$ and dim $H^1(\mathrm{UConf}_2(Y); \mathbb{Q}) = 2$, each licensed by [3] with the bound of [4]. The $K = 3$ rows are discrete-complex invariants only. Downgrading their interpretation does not downgrade the computation.

7. Verification

All linear algebra is over $\mathbb{Q}$ with exact rational arithmetic; no floating point is used and no rank is reported subject to a tolerance. The checks behind the statements above:

Statement	Check
Theorem 3.1, both directions	20 substrates at K = 2 and 3 (cycles, complete graphs, bull, butterfly, paw, $K_{3,3}$, Q_3, Petersen, Heawood, paths, stars, trees, C_5+chord, the 12-vertex witness of [1]): 0 mismatches between the criterion and the computed families.
Theorem 3.1, small cases	Exhaustive enumeration of all labeled simple graphs on 4 and 5 vertices containing a 4-cycle — 10 and 476 — at every $2 \le K \le n-1$. No violation.

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Statement	Check
Complementation preserves $\mathcal{K}$disj	Cell counts (|$\mathcal{K}_4$|, |$\mathcal{K}$disj|) at K and n–K agree exactly on Q_3 (K = 2…6), $K_{3,3}$, Petersen and C_7, confirming that the token-graph isomorphism of [2] carries commuting squares to commuting squares.
Corollary 4.1 and the bull control	ε harm computed under each predicate separately; the bull gives 3, 3, 0 for $\mathcal{K}$disj, $\mathcal{K}_4$, $\mathcal{K}_{3,4}$.
Proposition 5.1	Token graph built directly from the tree; H^1 by exact rank over $\mathbb{Q}$.
Table 2	Essential vertices and minimum essential-vertex path lengths computed by breadth-first search; girth by BFS.

Table 3. Verification protocol. Two independent implementations of the invariant cochain machinery, described in [1, §9.1], agree on every quantity they share with this paper.

8. Scope

Established. Theorem 3.1 in both directions for all $2 \le K \le n-1$: uniformly for $n \ge 6$, via the complementation of [2] and an explicit construction, and by exhaustive enumeration for the finite cases $n = 4$ and 5. Corollary 4.1 and the independence of its two hypotheses. Proposition 5.1. The Table 1 values as discrete configuration-space cohomology, and the Table 2 rows marked yes as continuous configuration-space cohomology.

Not established. Any statement for substrates, token numbers or groups outside the ranges tabulated. A continuous reading of the $K = 3$ rows. Abrams' theorem and the Prue–Scrimshaw improvement, which are cited with their hypotheses verified per graph and are not reproved here. Closed forms for the Table 1 values as functions of the substrate.

Relation to Paper I. Nothing in [1] is retracted. Its Lemma 6.2, stated for C_n with $n \ge 5$, is a true special case of Theorem 3.1 here, and its Theorem 6.3 is unaffected. One reading of [1, §8.4] should be guarded: the remark that the Prue–Scrimshaw improvement is irrelevant is correct as scoped there, namely for cycle substrates, and §6.1 above shows it does not extend — for every non-cycle substrate considered here the improvement is exactly what licenses the continuous reading.

References

[1] Kulik, D. *Rank access and closure depth in the relation-layer cohomology of token arenas.* QuHarmonics Research Group, 2026.

[2] Fabila-Monroy, R., Hurtado, F., Flores-Peñaloza, D., Huemer, C., Urrutia, J. and Wood, D. R. *Token graphs.* arXiv:0910.4774. The complementation $F_k(G) \cong F_{n-k}(G)$ is equation (2) of §2 (Basic Properties).

[3] Abrams, A. *Configuration spaces and braid groups of graphs.* PhD thesis, University of California, Berkeley, 2000.

[4] Prue, P. and Scrimshaw, T. *Abrams's stable equivalence for graph braid groups.* Topology and its Applications, 2014. arXiv:0909.5511.

[5] Abrams, A. and Ghrist, R. *Finding topology in a factory: configuration spaces.* American Mathematical Monthly 109 (2002), 140–150. arXiv:math/0009118.

[6] Farley, D. and Sabalka, L. *Discrete Morse theory and graph braid groups.* Algebraic & Geometric Topology 5 (2005), 1075–1109.

[7] Forman, R. *Morse theory for cell complexes.* Advances in Mathematics 134 (1998), 90–145.

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[8] An, B. H., Drummond-Cole, G. C. and Knudsen, B. *Edge stabilization in the homology of graph braid groups.* Geometry & Topology 24 (2020), 421–469.

Appendix. The instance set

The twenty substrates used to test Theorem 3.1, with their 4-cycle status and the resulting verdict. The set was chosen so that members *could* violate the criterion, which is why it contains triangle-bearing 4-cycle-free graphs and acyclic graphs as well as the expected positive and negative cases.

X	contains a 4-cycle	$\mathcal{K}_4$ vs $\mathcal{K}$disj
C_4	yes	≠
C_5, C_6, C_7	no	=
K_3 (triangle)	no	=
K_4, K_5	yes	≠
bull	no	=
butterfly	no	=
paw	no	=
$K_{3,3}$	yes	≠
Q_3 (cube)	yes	≠
Petersen	no	=
path P_5	no	=
star $K_{1,4}$	no	=
tree Y	no	=
12-vertex witness of [1]	yes	≠
two triangles joined by an edge	no	=
C_5 + chord	yes	≠
Heawood	no	=

Table 4. The instance set. Every row agrees with Theorem 3.1 at K = 2 and, where the token graph was small enough to enumerate, at K = 3.

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